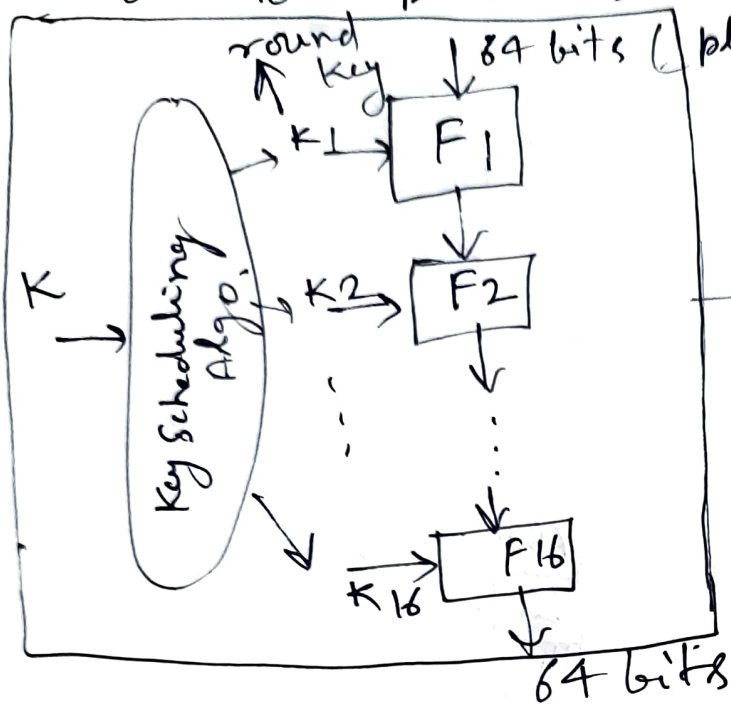
FIPS PUB 46 - Federal information processing standards ①

DES

No. of Rounds ⇒ 16.

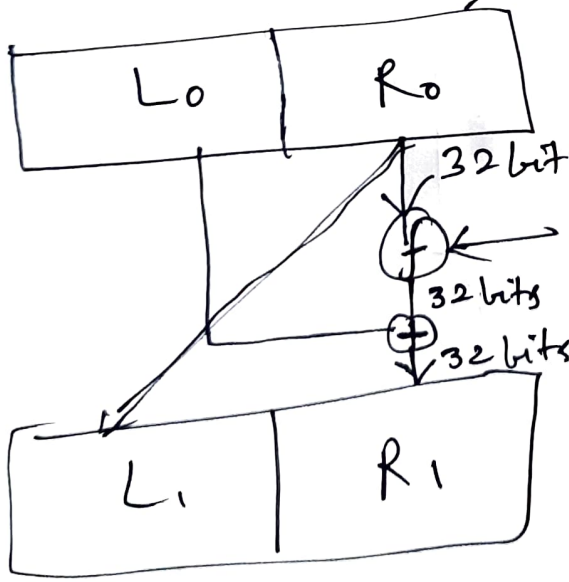
64-bit plaintext



→ Typical 16 rounds block cipher

DES Round Function : it's called Feistal cipher

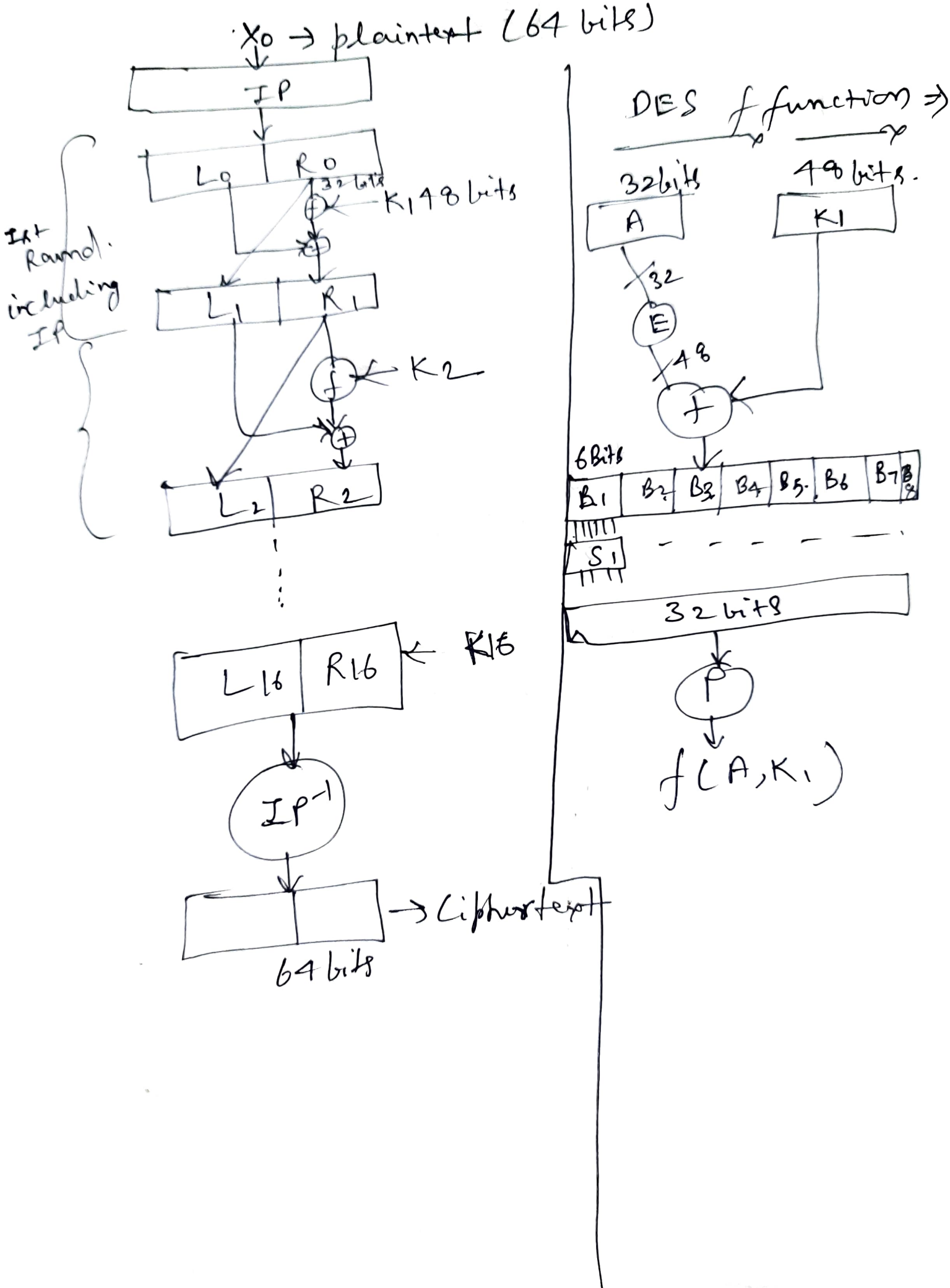
Feistal cipher → 64 bits input



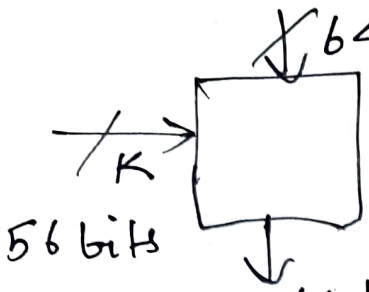
$L_1 = R_0$

$R_1 = f(R_0, K_1) \oplus L_0$

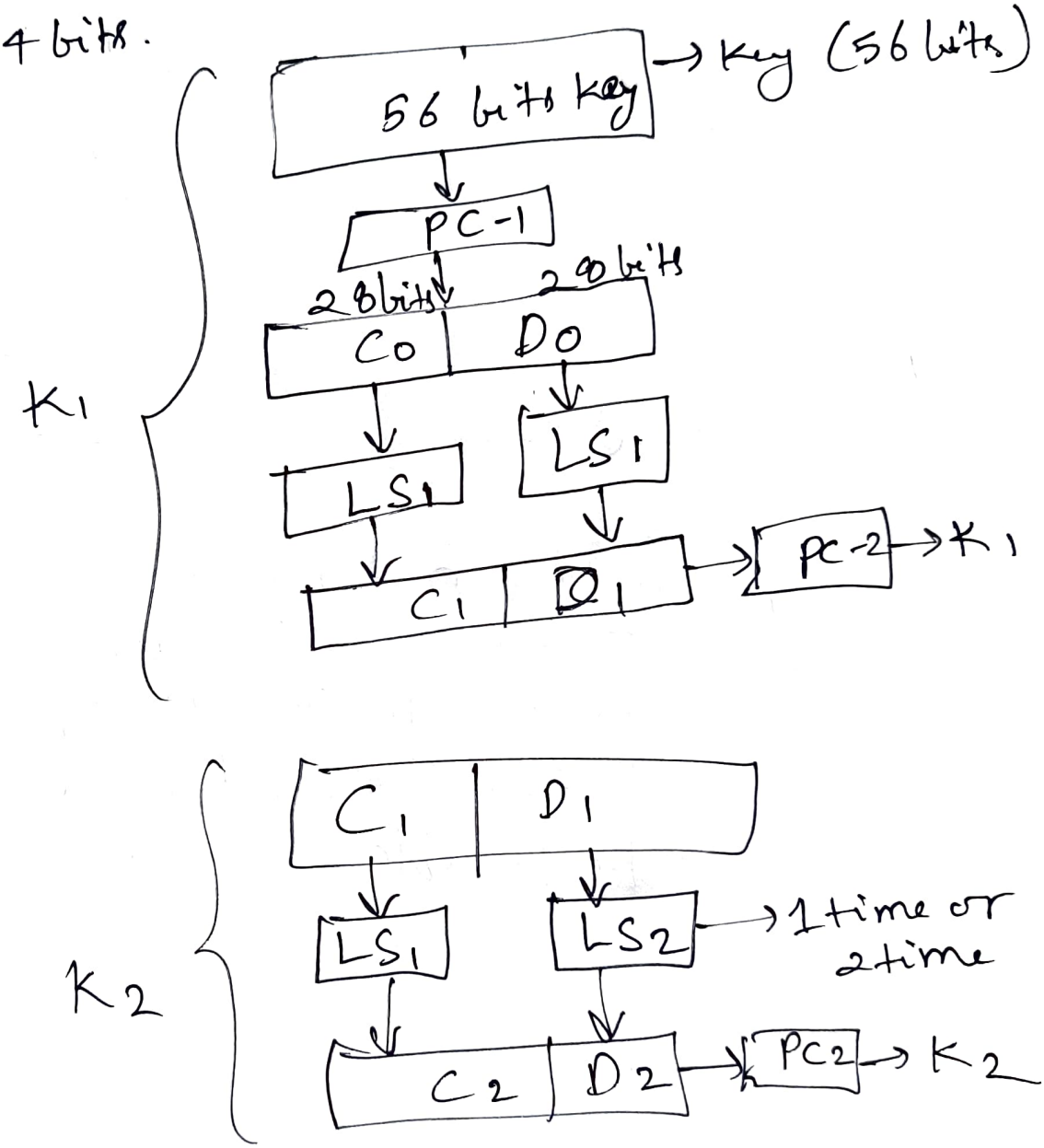
each round of DES is basically Feistal function.


X_0 → plaintext (64 bits)
IP
L_0
R_0
32 bits
K_1 48 bits
1st Round including IP
L_1
R_1
K_2
L_2
R_2
L_{16}
R_{16}
K_{16}
IP^{-1}
64 bits
→ Ciphertext
DES f function ⇒
32bits
48 bits.
A
K_1
32
E
48
6Bits
B_1 B_2 B_3 B_4 B_5 B_6 B_7 B_8
S_1
32 bits
P
$f(A, K_1)$

Key Scheduling for DES



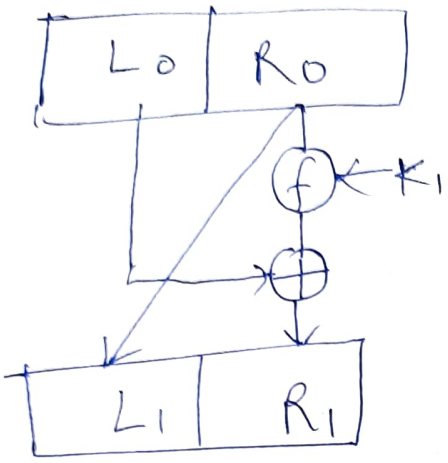
parity bits. →



K16

Invert Feistal cipher ? For Decryption

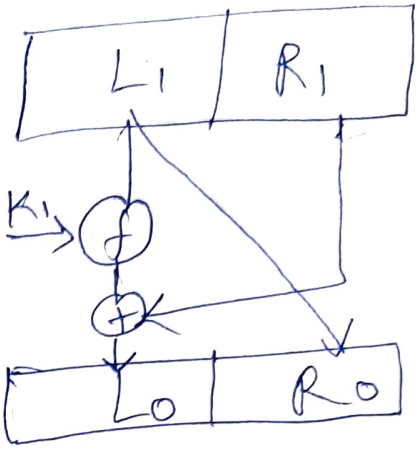
For Encryption (one Round)



$L_1 = R_0$

$R_1 = f(R_0, K_1) \oplus L_0$

For Decryption (one Round)



$R_0 = L_1$

$L_0 = R_1 \oplus f(R_0, K_1)$

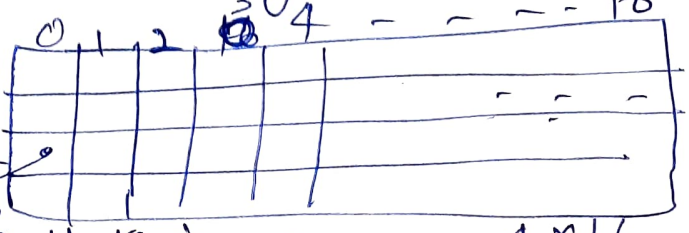
$= R_1 \oplus f(L_1, K_1)$

Beauty: We are not inverting f, we are applying same f in Decryption, Because sometimes inverse may not be possible, we have S box in some f.

How secure DES is?

Cryptanalysis on DES

Issues: (i) It is not well documented. For Example S-box, they have given just a table:

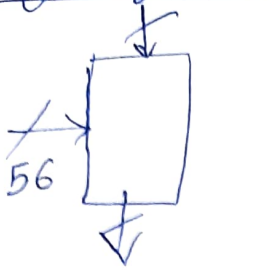


4×16

we have to do the table lookup.

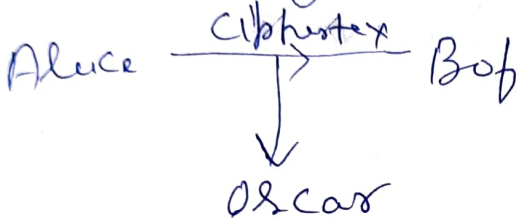
But How values are coming there is no proper operation for that. S,

(ii) Key size :⇒



Attack Model

(i) Ciphertext only attack:



(ii) Know plaintext attack:

attacker has some plaintext & corresponding ciphertext $\langle p_i, c_i \rangle$ $i = 1, \ldots, n$

goal :- to get the k

- to guess new plaintext p^* from c^*

(iii) Chosen Plaintext Attack: (temporary giving access to encryption system)

Attacker will chose some plaintext & get some ciphertext

goal :- get the key

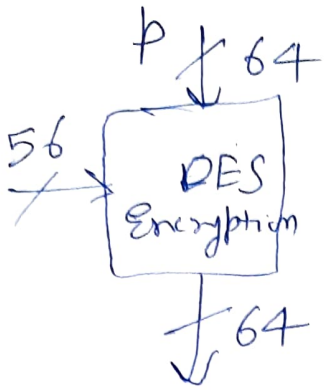
- or guess the p^* from c^*

(iv) Chosen ciphertext Attack: (temporary giving access to decryption system)

goal :- get the key

- or new plaintext p^* from c^*

Exhaustive Search Attack on DES



Alice ~~~~~ Bob.

$|K| = 56$

This is basically a know plaintext attack;

If Attacker knows p & its corresponding C.

$<P, C>$

- Attacker wants to guess the key

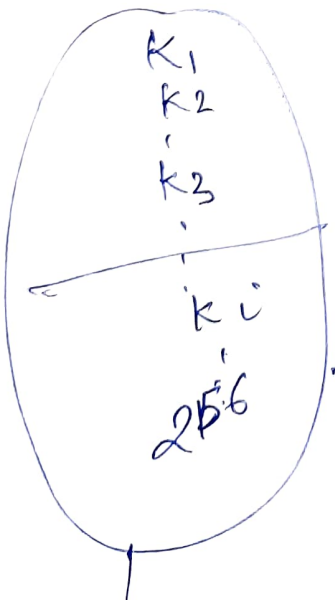
$K = 56$

Size of key space $= 2^{56}$

What attacker will do?

Attacker will try for all possible key, to decrypt the C.

& match p_i with p.----



Time complexity = ?

Time $= 2^{56} \times$ (time required for DES decryption or encryption)

$= 2^{56}$ sec.. 1 sec.

If two comb.s involved then time will be reduced by $\frac{2^{56}}{2} = 2^{55}$

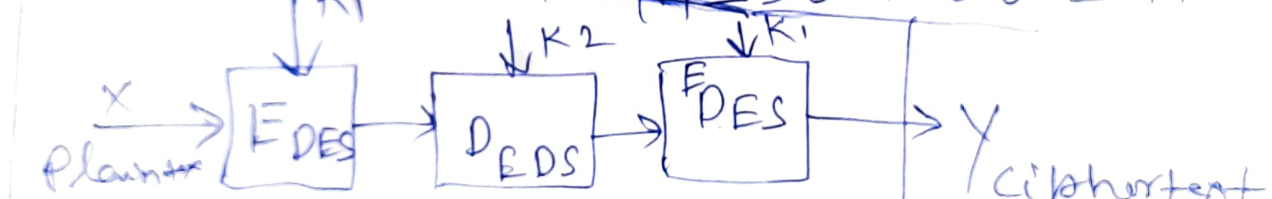
Exhaustive Search attack is know as generic attack, because attacker is not interested to know the internal structure of DES, only to know key size.

Triple DES

Replacement of DES ?

Encryption ⇒ $K = K_1 || K_2$

$|K| = 56 + 56 = 112$ bits.



↳ Encryption of triple DES

No. of Rounds = 48 rounds

↓

Major Drawback

↳ Huge time required.

Decryption ⇒ of Triple DES

